第四讲 自然演绎推理系统

1. 字符集

(1) 表示命题的符号：$p_1, p_2, \cdots, p_n, \cdots$

(2) 完备的联结词符号：$\{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$

(3) 辅助符号：( )

2. 命题公式

(1) 任意一个表示命题的符号为命题公式.

(2) 如果 A.B 是命题公式，则 $(\neg A)$, $(A \wedge B)$, $(A \vee B)$, $(A \rightarrow B)$, $(A \leftrightarrow B)$ 为命题公式.

(3) 有限次使用 (1) 与 (2) 所得到的结果均是命题公式.

3. 公理

$\Gamma, A \vdash A$ 其中 Γ 为公式集. Γ, A 表示 $\Gamma \cup \{A\}$

4. 推理规则

(1) 假设引入规则 (假设+)

$$\frac{\Gamma \vdash B}{\Gamma, A \vdash B}$$

(2) 假设消除规则 (假设−)

$$\frac{\Gamma, A \vdash B, \quad \Gamma, \neg A \vdash B}{\Gamma \vdash B.}$$

(3) $\vee$ 引入规则 ($\vee$+)

$$\frac{\Gamma \vdash A}{\Gamma \vdash A \vee B}, \quad \frac{\Gamma \vdash A}{\Gamma \vdash B \vee A.}$$

(4) $\vee$ 消除规则 ($\vee$−)

$$\frac{\Gamma, A \vdash C \ ; \ \Gamma, B \vdash C \ ; \ \Gamma \vdash A \vee B}{\Gamma \vdash C}$$

(5) $\wedge$ 引入规则 ($\wedge$+)

$$\frac{\Gamma \vdash A \ ; \ \Gamma \vdash B}{\Gamma \vdash A \wedge B}$$

(6) $\wedge$ 消除规则 ($\wedge$−)

$$\frac{\Gamma \vdash A \wedge B}{\Gamma \vdash A} \qquad \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash B}$$

(7) $\rightarrow$ 引入规则 ($\rightarrow$+)

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \rightarrow B}$$

(8) $\rightarrow$ 消除规则 ($\rightarrow$−)

$$\frac{\Gamma \vdash A \ ; \ \Gamma \vdash A \rightarrow B}{\Gamma \vdash B}$$

(9) ¬ 引入规则 (¬ +)

$$\frac{\Gamma, A \vdash B \qquad \Gamma, A \vdash \neg B}{\Gamma \vdash \neg A.}$$

(10) ¬ 消除规则 (¬ −)

$$\frac{\Gamma \vdash A \ ; \ \Gamma \vdash \neg A}{\Gamma \vdash B}$$

说明 ¬ 为假

$A \wedge \neg A \to B$

前提为假

结论为真

(11) ¬¬ 引入规则 (¬ ¬ +)

$$\frac{\Gamma \vdash A}{\Gamma \vdash \neg\neg A}$$

(12) ¬¬ 消除规则 (¬ ¬ −)

$$\frac{\Gamma \vdash \neg\neg A}{\Gamma \vdash A}$$

(13) ⟷ 引入规则 (⟷ +)

$$\frac{\Gamma \vdash A \to B \ ; \ \Gamma \vdash B \to A}{\Gamma \vdash A \leftrightarrow B}$$

(14) ⟷ 消除规则 (⟷ −).

$$\frac{\Gamma \vdash A \leftrightarrow B}{\Gamma \vdash A \to B \quad \Gamma \vdash B \to A.}$$

5. 定理推导.

约定：在ND中，以下序列称为 $\Gamma \vdash_{ND} A$ 的一个证明.

$\Gamma_1 \vdash A_1$, $\Gamma_2 \vdash A_2$, $\cdots$, $\Gamma_m \vdash A_m$ $(= \Gamma \vdash A)$ 其中 $\Gamma_i \vdash A_i$ $(i = 1, 2, \cdots, m)$

或为ND的公理，或为 $\Gamma_j \vdash A_j$ $(j < i)$ $\Gamma_{j_1} \vdash A_{j_1}$, $\Gamma_{j_2} \vdash A_{j_2}$, $\cdots$ $\Gamma_{j_k} \vdash A_{j_k}$

$(j_1, j_2, \cdots, j_k < i)$ 使用推理规则推出.

如果 $\Gamma = \{A\}$，则 $\Gamma \vdash B$ 简记为 $A \vdash B$.

如果 $\Gamma = \phi$，此时 $\Gamma \vdash A$，记为 $\vdash A$，称 A 为ND的定理.

- 定理：$\vdash A \vee \neg A$

证明：(1) $A \vdash A$ (公理)

(2) $A \vdash A \vee \neg A$ (1) ($\vee$ +)

(3) $\neg A \vdash \neg A$ (公理)

(4) $\neg A \vdash A \vee \neg A$ ($\vee$ +)

(5) $\vdash A \vee \neg A$ (2)(4) (假设 −)

- 定理：$\neg(A \vee B) \leftrightarrow \neg A \wedge \neg B$.

(1) $\neg(A \vee B), A \vdash A$ (公理)

(2) $\neg(A \vee B), A \vdash A \vee B$ (1)($\vee+$)

(3) $\neg(A \vee B), A \vdash \neg(A \vee B)$ (公理)

(4) $\neg(A \vee B) \vdash \neg A$ (2)(3)($\neg+$)

(5) $\neg(A \vee B), B \vdash B$ (公理)

(6) $\neg(A \vee B), B \vdash A \vee B$ (5)($\vee+$)

(7) $\neg(A \vee B), B \vdash \neg(A \vee B)$ (公理)

(8) $\neg(A \vee B) \vdash \neg B$ (6)(7)($\neg+$)

(9) $\neg(A \vee B) \vdash \neg A \wedge \neg B$ (4)(8)($\wedge+$)

(10) $\vdash \neg(A \vee B) \rightarrow \neg A \wedge \neg B$ (9)($\rightarrow+$)

(11) $\neg A \wedge \neg B, A \vee B \vdash \neg A \wedge \neg B$ (公理)

(12) $\neg A \wedge \neg B, A \vee B \vdash \neg A$ (11)($\wedge-$).

(13) $\neg A \wedge \neg B, A \vee B, A \vdash A$ (公理)

(14) $\neg A \wedge \neg B, A \vee B, B \vdash B$ (公理)

(15) $\neg A \wedge \neg B, A \vee B, B \vdash \neg A \wedge \neg B$ (公理)

(16) $\neg A \wedge \neg B, A \vee B, B \vdash \neg B$ (15)($\wedge-$)

(17) $\neg A \wedge \neg B, A \vee B, B \vdash A$ (14)(16)($\neg-$)

(18) $\neg A \wedge \neg B, A \vee B \vdash A \vee B$ (公理)

(19) $\neg A \wedge \neg B, A \vee B \vdash A$ (13)(17)(18)($\vee-$)

(20) $\neg A \wedge \neg B \vdash \neg(A \vee B)$ (12)(19)($\neg+$).

- 定理；$\neg(A \wedge B) \vdash \neg A \vee \neg B$

证明：

(1) $\neg(A \wedge B), \neg A \vdash \neg A$ (公理)

(2) $\neg(A \wedge B), \neg A \vdash \neg A \vee \neg B$ (1)($\vee+$)

(3) $\neg(A \wedge B), A, B \vdash A$ (公理)

(4) $\neg(A \wedge B), A, B \vdash B$ (公理)

(5) $\neg(A \wedge B), A, B \vdash A \wedge B$ (3)(4)($\wedge+$)

(6) $\neg(A \wedge B), A, B \vdash \neg(A \wedge B)$ (公理)

(7) $\neg(A \wedge B), A \vdash \neg B$ (5)(6)($\neg+$)

(8) $\neg(A \wedge B), A \vdash \neg A \vee \neg B$ (7)($\vee+$)

(9) $\neg(A \wedge B) \vdash \neg A \vee \neg B$ (2)(8)(假设-)

- 定理：$\neg A \vee \neg B \vdash \neg(A \wedge B)$

证明：

(1) $\neg A \vee \neg B, A \wedge B \vdash A \wedge B$ (公理)

(2) $\neg A \vee \neg B, A \wedge B \vdash A$ ($\wedge-$)

(3) $\neg A \vee \neg B, A \wedge B \vdash B$ ($\wedge-$)

(4) $\neg A \vee \neg B, A \wedge B, \neg A \vdash \neg A$ (公理)

(5) $\neg A \vee \neg B, A \wedge B, \neg B \vdash B$ (3)(假设+).

(6) $\neg A \vee \neg B, A \wedge B, \neg B \vdash \neg B$ (公理)

(7) $\neg A \vee \neg B, A \wedge B, \neg B \vdash \neg A$ (5)(6)($\neg-$)

(8) $\neg A \vee \neg B,\ A \wedge B \vdash \neg A \vee \neg B$ (公理).

(9) $\neg A \vee \neg B,\ A \wedge B \vdash \neg A$ (4)(7)(8)($\vee -$)

(10) $\neg A \vee \neg B \vdash \neg(A \vee B)$ (2)(9)($\neg +$).

- 定义：设A.B为任意两个命题公式，如果 $A \vdash B$，$B \vdash A$，则称A与B演绎等价.记为 $A \dashv\vdash B$

- 定理：$A \to B \vdash \neg \neg A \vee B$.

证明：先证 $A \to B \vdash A \vee B$ (1) $A \to B,\ A \vdash A \to B$ (公理)

(2) $A \to B,\ A \vdash A$ (公理)

(3) $A \to B,\ A \vdash B$ (1)(2)($\to -$)

(4) $A \to B,\ A \vdash \neg A \vee B$ (3)($\vee +$)

(5) $A \to B,\ \neg A \vdash \neg A$ (公理)

(6) $A \to B,\ \neg A \vdash \neg A \vee B$ (5)($\vee +$).

(7) $A \to B \vdash \neg A \vee B$ (4)(6)(假设-).

再证 $\neg A \vee B \vdash A \to B$

(8) $\neg A \vee B,\ A\ B \vdash B$. (公理) (12) $\neg A \vee B,\ A. \vdash \neg A \vee B$ (公理)

(9) $\neg A \vee B,\ A,\ \neg A \vdash \neg A$ (公理) (13) $\neg A \vee B,\ A. \vdash B$ (8)(11)(12)($\vee -$)

(10) $\neg A \vee B,\ A,\ \neg A \vdash A$ (公理) (14) $\neg A \vee B \vdash A \to B$ (13)($\to +$)

(11) $\neg A \vee B,\ A\ \neg A \vdash B$ (9)(10)($\vee -$).

- 定理：$A \to (B \to A)$

证明：(1) $A,\ B \vdash A$ (公理)

(2) $A \vdash B \to A$ (1)($\to +$).

(3) $\vdash A \to (B \to A)$ (2)($\to +$)

- 定理：$A \wedge (B \vee C) \vdash (A \wedge B) \vee (A \wedge C)$

证明：(1) $A \wedge (B \vee C) \vdash A \wedge (B \vee C)$ (公理)

(2) $A \wedge (B \vee C) \vdash A$ (1) $(\wedge -)$

(3) $A \wedge (B \vee C) \vdash B \vee C$ (1) $(\wedge -)$

(4) $A \wedge (B \vee C), B \vdash B$ (公理).

(5) $A \wedge (B \vee C), B \vdash A$ (2)(假设+).

(6) $A \wedge (B \vee C), B \vdash A \wedge B$ (4)(5) $(\wedge +)$.

(7) $A \wedge (B \vee C), B \vdash (A \wedge B) \vee (A \wedge C)$ (6) $(\vee +)$.

(8) $A \wedge (B \vee C), C \vdash C$ (公理)

(9) $A \wedge (B \vee C), C \vdash A$ (假设+)

(10) $A \wedge (B \vee C), C \vdash A \wedge C$ (8)(9) $(\wedge +)$

(11) $A \wedge (B \vee C), C \vdash (A \wedge B) \vee (A \wedge C)$ (10) $(\vee +)$.

(12) $A \wedge (B \vee C) \vdash (A \wedge B) \vee (A \wedge C)$ (7)(11)(3) $(\vee -)$.

- 定理：$(A \to (B \to C)) \to ((A \to B) \to (A \to C))$

证明：(1) $A \to (B \to C), A \to B, A \vdash A$ (公理)

(2) $A \to (B \to C), A \to B, A \vdash A \to (B \to C)$ (公理).

(3) $A \to (B \to C), A \to B, A \vdash B \to C$ (1)(2) $(\to -)$

(4) $A \to (B \to C), A \to B, A \vdash A \to B$ (公理)

(5) $A \to (B \to C), A \to B, A \vdash B$ (1)(4) $(\to -)$

(6) $A \to (B \to C), A \to B, A \vdash C$ (3)(5) $(\to -)$

(7) $A \to (B \to C), A \to B \vdash A \to C$ (6) $(\to +)$

(8) $A \to (B \to C) \vdash (A \to B) \to (A \to C)$ (7) $(\to +)$

(9) $\vdash (A \to (B \to C)) \to ((A \to B) \to (A \to C))$ (8) $(\to +)$

- 定理：$\vdash (\neg A \to \neg B) \to (B \to A)$

证明：(1) $\neg A \to \neg B, B, \neg A \vdash \neg A$ (公理)

(2) $\neg A \to \neg B, B, \neg A \vdash \neg A \to \neg B$ (公理).

(3) $\neg A \to \neg B, B, \neg A \vdash \neg B$ (1)(2) $(\to -)$

(4) $\neg A \to \neg B, B, \neg A \vdash B$ (公理)

(5) $\neg A \to \neg B, B \vdash \neg \neg A$ (3)(4) $(\neg +)$

(6) $\neg A \to \neg B, B \vdash A$ (5) $(\neg \neg -)$

(7) $\neg A \to \neg B \vdash B \to A$ (6) $(\to +)$

(8) $\vdash (\neg A \to \neg B) \to (B \to A)$ (7) $(\to +)$